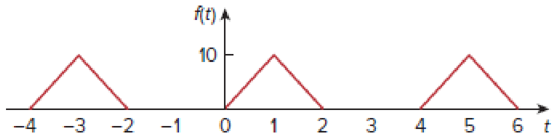
Chapter 17, Problem 19P

Problem

Obtain the Fourier series for the periodic waveform in Fig. 17.57.

Figure. 17.57:



Step-by-step solution

Step 1 of 8

Refer to waveform in Figure 17.57 in the textbook.

The time period T is 4.

The angular frequency is,

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$$

Let us assume that the amplitude of waveform is 10.

Write the mathematical expressions of the Figure 1.

$$f(t) = \begin{cases} 10t & \text{for } 0 < t < 1 \\ 10(2-t) & \text{for } 1 < t < 2 \end{cases}$$

Step 2 of 8

Calculate the value of a_0.

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt$$
$$= \frac{1}{4}\int_0^1 10t\,dt + \frac{1}{4}\int_1^2 10(2-t)\,dt$$
$$= \frac{10}{4}\frac{t^2}{2}\Big|_0^1 + \frac{10}{4}\left[2t - \frac{t^2}{2}\right]_1^2$$
$$= \frac{5}{4}(1-0) + \frac{10}{4}\left\{\left[2(2) - \frac{(2)^2}{2}\right] - \left[2(1) - \frac{(1)^2}{2}\right]\right\}$$
$$= \frac{5}{4} + \frac{5}{2}\left(\frac{1}{2}\right)$$
$$= 2.5$$

Step 3 of 8

Calculate the value of a_n.

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t dt$$

$$= \left\{\begin{array}{l} \frac{2}{4}\int_0^1 10t\cos n\omega_0 tdt + \\ \frac{2}{4}\int_1^2 10(2-t)\cos n\omega_0 tdt \end{array}\right\} \qquad \left[\text{since} \int fd(g)dt = fg - \int d(f)g\right]$$

$$= \left\{\begin{array}{l} \frac{20}{4}\left[\frac{1}{(n\omega_0)}t\sin n\omega_0 t + \frac{1}{(n\omega_0)^2}\cos n\omega_0 t\right]_0^1 + \\ \frac{40}{4}\int_1^2 \cos n\omega_0 tdt - \frac{20}{4}\int_1^2 t\cos n\omega_0 tdt \end{array}\right\}$$

$$= \left\{\begin{array}{l} \frac{5}{(n\omega_0)^2}\left[(n\omega_0)t\sin n\omega_0 t + \cos n\omega_0 t\right]_0^1 + \frac{10}{(n\omega_0)}\sin n\omega_0 t\Big|_1^2 \\ -5\left[\frac{1}{(n\omega_0)}t\sin n\omega_0 t + \frac{1}{(n\omega_0)^2}\cos n\omega_0 t\right]_1^2 \end{array}\right\}$$

Step 4 of 8

Simplify further.

$$a_n = \left\{\begin{array}{l} \frac{5}{(n\omega_0)^2}\left[(n\omega_0)\sin n\omega_0 + \cos n\omega_0 - 1\right] + \frac{10}{(n\omega_0)}\left[\sin 2n\omega_0 - \sin n\omega_0\right] \\ -\frac{5}{(n\omega_0)^2}\left[2(n\omega_0)\sin 2n\omega_0 + \cos 2n\omega_0 - (n\omega_0)\sin n\omega_0 - \cos n\omega_0\right] \end{array}\right\}$$

$$= \left\{\begin{array}{l} \frac{5}{(n\omega_0)^2}\left[2(n\omega_0)\sin n\omega_0 - 2(n\omega_0)\sin 2n\omega_0 + 2\cos n\omega_0 - \cos 2n\omega_0 - 1\right] \\ +\frac{10}{(n\omega_0)}\left[\sin 2n\omega_0 - \sin n\omega_0\right] \end{array}\right\}$$

$$= \frac{5}{(n\omega_0)^2}\left\{\begin{array}{l} 2(n\omega_0)\sin n\omega_0 - 2(n\omega_0)\sin 2n\omega_0 + 2\cos n\omega_0 - \\ 2\sin n\omega_0 + 2\sin 2n\omega_0 - \cos 2n\omega_0 - 1 \end{array}\right\} \qquad \left(\text{since } \omega_0 = \frac{\pi}{2}\right)$$

$$= \frac{5}{\left(n\frac{\pi}{2}\right)^2}\left\{\begin{array}{l} 2\left(n\frac{\pi}{2}\right)\sin n\frac{\pi}{2} - 2\left(n\frac{\pi}{2}\right)\sin 2n\frac{\pi}{2} + 2\cos n\frac{\pi}{2} - \\ 2\sin n\frac{\pi}{2} + 2\sin 2n\frac{\pi}{2} - \cos 2n\frac{\pi}{2} - 1 \end{array}\right\} \left(\begin{array}{l} \text{since } \sin n\pi = 0 \\ \cos n\pi = (-1)^n \end{array}\right)$$

$$= \frac{20}{(n\pi)^2}\left\{(n\pi)\sin\frac{n\pi}{2} - (n\pi)\sin n\pi + \cos\frac{n\pi}{2} - 4\sin\frac{n\pi}{2} - 1\right\}$$

$$= \begin{cases} \frac{20}{(n\pi)^2}\left[(-1)^{\frac{n-1}{2}}[n\pi - 4] - 1\right], & n = \text{odd} \\ \frac{20}{(n\pi)^2}\left[(-1)^{\frac{n}{2}} - 1\right], & n = \text{even} \end{cases}$$

Step 5 of 8

Calculate the value of b_n.

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t dt$$

$$= \begin{Bmatrix} \frac{2}{4}\int_0^1 10t \sin n\omega_0 t dt + \\ \frac{2}{4}\int_1^2 10(2-t)\sin n\omega_0 t dt \end{Bmatrix} \qquad \left[\text{since} \int f d(g) dt = fg - \int d(f) g\right]$$

$$= \begin{Bmatrix} \frac{20}{4}\left[-\frac{1}{(n\omega_0)} t\cos n\omega_0 t + \frac{1}{(n\omega_0)^2}\sin n\omega_0 t\right]\Bigg|_0^1 + \\ \frac{40}{4}\int_1^2 \sin n\omega_0 t dt - \frac{20}{4}\int_1^2 t\sin n\omega_0 t dt \end{Bmatrix}$$

$$= \begin{Bmatrix} \frac{5}{(n\omega_0)^2}\left[-(n\omega_0) t\cos n\omega_0 t + \sin n\omega_0 t\right]\Big|_0^1 - \frac{10}{(n\omega_0)}\cos n\omega_0 t\Big|_1^2 \\ -5\left[-\frac{1}{(n\omega_0)} t\cos n\omega_0 t + \frac{1}{(n\omega_0)^2}\sin n\omega_0 t\right]_1^2 \end{Bmatrix}$$

Step 6 of 8

Simplify further.

$$b_n = \begin{Bmatrix} \frac{5}{(n\omega_0)^2}\left[-(n\omega_0)\cos n\omega_0 + \sin n\omega_0\right] - \frac{10}{(n\omega_0)}\left[\cos 2n\omega_0 - \cos n\omega_0\right] \\ \frac{5}{(n\omega_0)^2}\left[2(n\omega_0)\cos 2n\omega_0 - \sin 2n\omega_0 + (n\omega_0)\cos n\omega_0 + \sin n\omega_0\right] \end{Bmatrix}$$

$$= \begin{Bmatrix} \frac{5}{(n\omega_0)^2}\begin{bmatrix} -(n\omega_0)\cos n\omega_0 + \sin n\omega_0 + 2(n\omega_0)\cos 2n\omega_0 - \sin 2n\omega_0 \\ +(n\omega_0)\cos n\omega_0 + \sin n\omega_0 \end{bmatrix} \\ -\frac{10}{(n\omega_0)}\left[\cos 2n\omega_0 - \cos n\omega_0\right] \end{Bmatrix}$$

$$= \frac{5}{(n\omega_0)^2}\begin{Bmatrix} -(n\omega_0)\cos n\omega_0 + \sin n\omega_0 + 2(n\omega_0)\cos 2n\omega_0 - \\ \sin 2n\omega_0 + (n\omega_0)\cos n\omega_0 + \sin n\omega_0 \\ -2\cos 2n\omega_0 + 2\cos n\omega_0 \end{Bmatrix}$$

$$= \frac{5}{(n\omega_0)^2}\begin{Bmatrix} 2\sin n\omega_0 + 2(n\omega_0)\cos 2n\omega_0 - \\ \sin 2n\omega_0 - 2\cos 2n\omega_0 + 2\cos n\omega_0 \end{Bmatrix} \qquad \left(\text{since } \omega_0 = \frac{\pi}{2}\right)$$

Step 7 of 8

Simplify further.

$$b_n = \frac{5}{\left(n\frac{\pi}{2}\right)^2}\begin{Bmatrix} 2\sin n\frac{\pi}{2} + 2\left(n\frac{\pi}{2}\right)\cos 2n\frac{\pi}{2} - \\ \sin 2n\frac{\pi}{2} - 2\cos 2n\frac{\pi}{2} + 2\cos n\frac{\pi}{2} \end{Bmatrix}$$

$$= \frac{20}{(n\pi)^2}\begin{Bmatrix} 2\sin\frac{n\pi}{2} + n\pi\cos n\pi - \\ \sin n\pi - 2\cos n\pi + 2\cos\frac{n\pi}{2} \end{Bmatrix}$$

$$= \begin{cases} \frac{20}{(n\pi)^2}\left[2(-1)^{\frac{n-1}{2}} + (-1)^n\left[n\pi - 2\right]\right], & n = \text{odd} \\ \frac{20}{(n\pi)^2}\left[(-1)^n\left[n\pi - 2\right] + 2(-1)^{\frac{n}{2}}\right], & n = \text{even} \end{cases}$$

Write the expression for Fourier series.

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t \right)$$

For $n = \text{odd}$,

$$f(t) = 2.5 + \frac{20}{(n\pi)^2} \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left\{ \begin{array}{l} \left[(-1)^{\frac{n-1}{2}} [n\pi - 4] - 1 \right] \cos \frac{n\pi}{2} t + \\ \left[(-1)^n [n\pi - 2] + 2(-1)^{\frac{n-1}{2}} \right] \sin \frac{n\pi}{2} t \end{array} \right\}$$

Therefore, Fourier series, $f(t)$ for the periodic waveform in Figure 1 is,

$$\boxed{2.5 + \frac{20}{(n\pi)^2} \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left\{ \begin{array}{l} \left[(-1)^{\frac{n-1}{2}} [n\pi - 4] - 1 \right] \cos \frac{n\pi}{2} t + \\ \left[(-1)^n [n\pi - 2] + 2(-1)^{\frac{n-1}{2}} \right] \sin \frac{n\pi}{2} t \end{array} \right\}}.$$